

Micro-Internship Portal For Computer Department MITAOE Students

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Abstract—Early-year engineering students often face difficulty in obtaining practical industry exposure due to the limited availability of short-term and skill-appropriate internship opportunities. Conventional internship platforms are primarily designed for senior students and lack institutional verification and academic alignment.

This paper presents a secure, institute-driven Micro-Internship Portal that connects students, organizations, and faculty administrators through a centralized web-based platform. The system enables organizations to post short-duration, task-based micro-internships, while students can apply based on their skills and academic level. Faculty administrators verify organizations and internship postings to ensure authenticity and relevance. The proposed architecture uses a modern web stack with role-based access control, real-time data management, and cloud deployment. This approach improves early career exposure, trust, and industry-academia collaboration.

Index Terms—Micro-internships, Experiential learning, Web-based systems, Industry-academia collaboration, Student skill development

I. INTRODUCTION

Engineering education traditionally emphasizes theoretical knowledge, with practical exposure often introduced in later stages of the academic curriculum. However, the rapidly evolving technological landscape demands that students develop hands-on skills and real-world problem-solving abilities early in their academic journey. Employers increasingly seek graduates who are not only technically proficient but also capable of applying their knowledge in real-world scenarios.

Despite this demand, early-year engineering students face significant challenges in accessing relevant internship opportunities. Conventional internship platforms primarily focus on long-duration roles targeted toward senior students, making them unsuitable for beginners. As a result, students often rely on informal and unverified sources, leading to confusion, limited exposure, and a lack of structured guidance.

Micro-internships have emerged as an effective solution to this problem. These are short-duration, task-based opportunities that allow students to gain practical experience without

requiring advanced expertise or long-term commitments. They provide early exposure to industry workflows while enabling organizations to efficiently outsource small but meaningful tasks.

However, existing platforms lack critical features such as institutional verification, academic alignment, and personalized guidance tailored for early-stage learners. To address these limitations, this paper proposes an institute-driven Micro-Internship Portal that integrates students, organizations, and faculty administrators into a unified system.

Unlike traditional platforms, the proposed system incorporates a modern web architecture with a secure backend, real-time data synchronization, and an AI-based recommendation engine. The system uses a hybrid approach combining rule-based filtering and AI inference to generate match scores, identify relevant skills, and provide explainable recommendations. Additionally, a fallback mechanism ensures continuous functionality even when AI-based results are unavailable.

By combining institutional oversight, intelligent recommendations, and scalable system design, the proposed platform enhances early career exposure, improves decision-making, and strengthens industry-academia collaboration.

II. RELATED WORK

In recent years, the concept of experiential learning has gained significant attention as a means to bridge the gap between academic education and industry requirements. Several studies and platforms have explored alternative internship models that emphasize flexibility, skill-based learning, and early exposure to real-world problems.

One of the most prominent implementations of micro-internships is the Parker Dewey platform, which introduced short-duration, professional assignments aimed at helping students gain practical experience without long-term commitments. Their model demonstrates that task-based engagements can enhance student confidence, employability, and career

awareness, particularly for students who lack prior industry exposure. However, such platforms operate independently of academic institutions and do not provide faculty-level verification or academic alignment.

Universities have also begun experimenting with structured micro-internship programs. Adelphi University implemented an institution-supported micro-internship framework through its career development center, allowing students to work on short-term projects with external organizations. This initiative highlighted the effectiveness of guided, short-duration internships in improving student readiness and engagement. While successful, these implementations remain limited to specific institutions and lack scalable, technology-driven platforms that can be easily adapted by other colleges.

Research in work-based and experiential learning further supports the adoption of micro-internship models. Boud and Solomon emphasized that integrating workplace tasks with academic learning significantly enhances knowledge retention and practical understanding. Similarly, Eraut identified informal learning through real-world activities as a critical component of professional skill development. These studies underline the importance of hands-on experience in engineering education but do not address the technological infrastructure required to manage such opportunities at scale.

Several online internship platforms such as Internshala, LinkedIn, and Naukri provide access to a wide range of internship opportunities. However, these platforms are primarily designed for senior students or graduates and focus on long-term internships. They often lack verification mechanisms, guidance for early-year students, and academic oversight, making them less suitable for beginners seeking structured learning experiences.

From a system design perspective, previous research on web-based learning and recruitment systems highlights the importance of role-based access control, secure authentication, and scalable cloud-based architectures. While these systems demonstrate effective use of modern web technologies, they are not specifically tailored to micro-internship workflows or institute-driven verification processes.

Based on the review of existing platforms and academic literature, it is evident that there is a gap in providing a secure, institute-specific, and student-guided micro-internship system. Most existing solutions either focus on large-scale commercial recruitment or lack academic integration. The proposed Micro-Internship Portal addresses these gaps by combining experiential learning principles with an institute-driven, verified, and technology-enabled platform designed specifically for early-year engineering students.

III. PROBLEM STATEMENT AND MOTIVATION

Early-year engineering students face significant challenges in accessing meaningful industry exposure due to the lack of suitable internship opportunities aligned with their academic level and skill set. Most existing internship platforms primarily

TABLE I
COMPARISON WITH EXISTING INTERNSHIP PLATFORMS

Feature	Internshala	LinkedIn	Proposed System
Short-Term Tasks	Limited	No	Yes
Institute Verification	No	No	Yes
Faculty Monitoring	No	No	Yes
Beginner Friendly	Partial	No	Yes
Skill-Based Filtering	Yes	Yes	Yes

target senior students and graduates, offering long-term internships that require advanced technical expertise. As a result, early-year students often rely on informal sources such as social media, peer recommendations, or unverified postings, which leads to confusion and unreliable opportunities.

Organizations, particularly startups and small enterprises, frequently possess short-term tasks that do not justify hiring full-time interns. However, they lack a structured platform to connect with students willing to undertake such tasks. In parallel, academic institutions do not have centralized systems that allow faculty members to verify organizations, monitor internship activities, or ensure academic relevance.

The absence of a trusted, institute-level platform results in missed learning opportunities for students, inefficient task allocation for organizations, and lack of oversight for faculty. This motivates the need for a secure, verified, and institute-driven micro-internship system that bridges the gap between students, organizations, and academia.

IV. PROPOSED SYSTEM

A key enhancement in the updated system is the integration of an AI-based recommendation engine. The system evaluates student profiles and internship requirements using a hybrid approach combining rule-based filtering and external AI inference. The model generates match scores, identifies relevant skills, and provides reasoning, improving decision-making and user experience.

This paper proposes an institute-driven Micro-Internship Portal designed to facilitate short-duration, skill-based internships for early-year engineering students. The system provides a centralized platform where students, organizations, and faculty administrators interact through role-based access.

Students can create profiles, browse verified micro-internship opportunities based on required skills and domains, apply for tasks, and track application status. Organizations can register, submit internship tasks with defined deliverables, and review student applications. Faculty administrators act as system moderators, verifying organization authenticity and approving internship postings before they become visible to students.

The system emphasizes security, transparency, and academic alignment. Role-based authentication ensures controlled access, while administrative verification prevents fraudulent postings. By integrating experiential learning principles with

a structured digital workflow, the proposed system improves early industry exposure and strengthens industry–academia collaboration.

A. Student Module

The Student module is designed to support early-year engineering students in discovering and applying for suitable micro-internship opportunities. Students can register and create profiles containing academic details, skills, and areas of interest. Based on this information, the system displays verified micro-internships relevant to the student’s skill level and domain.

Students can browse internships using filters such as required skills, duration, and domain. Once an application is submitted, students can track its status in real time. This structured workflow ensures transparency and reduces confusion commonly faced while using unverified platforms.

B. Organization Module

The Organization module enables companies, startups, and professionals to post short-duration, task-based micro-internships. Organizations provide task descriptions, required skills, expected deliverables, and duration details.

Submitted internship postings are not immediately visible to students. Instead, they undergo administrative verification to ensure authenticity and relevance. Organizations can review student applications, shortlist candidates, and update application status through the system dashboard.

C. Administrator Module

The Administrator module is managed by faculty members who act as system moderators. Administrators verify organization registrations, approve or reject internship postings, and monitor system activity.

This module ensures academic alignment, platform security, and quality control. By involving faculty in the verification process, the system establishes trust among students and prevents fraudulent or irrelevant internship postings.

The key contributions of this work include:

- An institute-driven micro-internship platform with faculty verification
- A hybrid AI-based recommendation system with explainable outputs
- A real-time architecture using Firebase and REST APIs
- A fallback-enabled recommendation pipeline ensuring system reliability

V. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

The design of the Micro-Internship Portal is guided by a set of functional and non-functional requirements to ensure usability, security, and reliability.

A. Functional Requirements

The system shall allow students to register and create profiles containing academic details and skills. The system shall enable students to browse and apply for verified micro-internship opportunities. The system shall allow organizations to register and submit internship postings with defined requirements and deliverables. The system shall provide faculty administrators with the ability to verify organizations and approve or reject internship postings. The system shall support application tracking and status notifications for students and organizations.

B. Non-Functional Requirements

The system shall ensure secure authentication and role-based access control. The system shall be scalable to support a growing number of users and internship postings. The system shall provide real-time updates for application status and notifications. The system shall maintain data integrity and prevent unauthorized access. The system shall offer a responsive and user-friendly interface accessible across devices.

VI. SYSTEM ARCHITECTURE AND DESIGN

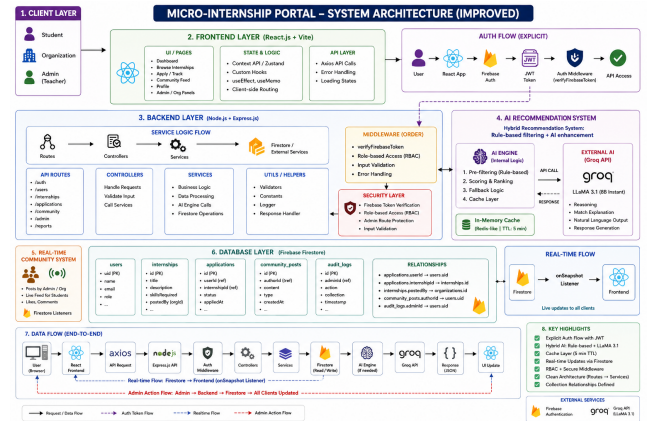


Fig. 1. System Architecture of the Institute-Driven Micro-Internship Portal

Figure 1 illustrates the layered architecture of the proposed system. The separation of concerns across frontend, backend, authentication, and database layers improves maintainability, scalability, and security. Each user interaction flows through controlled interfaces, ensuring role-based access and data integrity.

The Micro-Internship Portal follows a layered web-based architecture to ensure modularity, scalability, and security. The system consists of a client layer, frontend layer, authentication layer, backend layer, and database layer.

An additional AI layer is introduced in the architecture, which processes internship recommendations using external APIs. The system also includes caching and fallback mechanisms to ensure reliability and reduce latency.

The client layer includes three primary users: students, organizations, and faculty administrators. Users interact with the system through a React-based frontend that provides dashboards for browsing internships, posting tasks, and verifying applications.

Authentication is managed using Firebase Authentication, enabling secure login and role-based access control. The backend layer exposes RESTful APIs responsible for handling business logic such as internship posting, application processing, verification workflows, and notifications.

Data is stored in a cloud-hosted NoSQL database that maintains user profiles, internship listings, application records, and system logs. The architecture supports real-time updates, ensuring immediate status notifications to users. Cloud deployment and CI/CD integration further enhance reliability and scalability.

VII. SYSTEM DATA MODEL AND DATABASE DESIGN

A well-structured data model is essential for supporting the dynamic workflows of the Micro-Internship Portal. The system uses a NoSQL-based document-oriented database to efficiently manage heterogeneous data such as user profiles, internship postings, applications, and verification records.

A. Entity Description

The primary entities in the system include *Student*, *Organization*, *Administrator*, *Internship*, and *Application*. Each entity is stored as a separate collection to ensure modularity and scalability.

Student entities store academic information, skill sets, and application history. Organization entities maintain verification status, posted internships, and selection records. Internship entities include task descriptions, required skills, duration, and approval status. Application entities map students to internships and track selection progress.

B. Relationship Mapping

The data model supports one-to-many and many-to-many relationships. A single organization can post multiple internships, while each internship can receive multiple student applications. Faculty administrators maintain verification relationships with organizations and internship postings.

This flexible schema design enables efficient querying, real-time updates, and future feature expansion such as analytics and recommendation engines.

TABLE II
KEY DATABASE COLLECTIONS AND ATTRIBUTES

Collection	Key Attributes
Student	ID, Name, Skills, Year, Applications
Organization	ID, Name, Verification Status, Internships
Internship	ID, Skills Required, Duration, Approval Status
Application	Student ID, Internship ID, Status
Administrator	ID, Role, Verification Logs

VIII. SYSTEM WORKFLOW AND PROCESS ANALYSIS

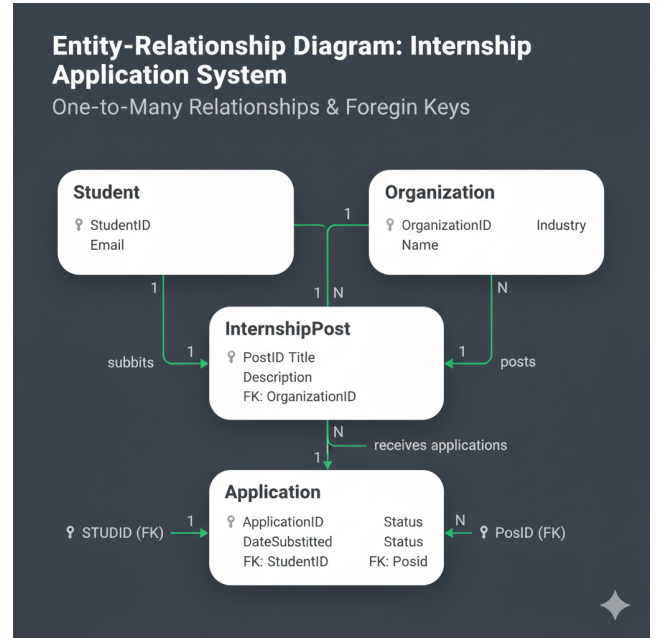


Fig. 2. Entity Relationship Diagram of the Micro-Internship Portal

Figure 2 illustrates the entity relationship model of the proposed system, showing core entities and their interactions.

The Micro-Internship Portal follows a structured workflow that ensures transparency, verification, and smooth interaction among all stakeholders.

Initially, users register on the platform based on their role. Students complete their profiles and explore available micro-internship opportunities. Organizations submit internship postings that include task descriptions, required skills, and expected outcomes.

Before being published, all internship postings undergo administrative verification. Faculty administrators review organization details and task relevance to ensure authenticity and academic alignment. Only approved postings are made visible to students.

Students apply for suitable internships, after which organizations review applications and update selection status. Notifications are generated at each stage to keep users informed. This workflow ensures a controlled, secure, and efficient micro-internship process while maintaining institutional oversight.

IX. USE CASE ANALYSIS

The system supports multiple use cases based on user roles.

A. Student Use Case

A student registers, logs in, browses verified micro-internships, applies for suitable tasks, and tracks application

status. Notifications inform the student about acceptance or rejection.

B. Organization Use Case

An organization registers on the platform, submits internship tasks, reviews applications, and selects suitable candidates after administrative approval.

C. Administrator Use Case

The administrator verifies organizations, approves internship postings, monitors system usage, and ensures academic relevance and platform integrity.

X. IMPLEMENTATION DETAILS

The system integrates external AI services using Groq API (LLaMA 3.1 model) for intelligent internship recommendations. Backend APIs handle AI requests, while fallback mechanisms ensure system continuity during API failures.

A. Frontend Implementation

The frontend is implemented using React.js, enabling a responsive and modular user interface. Components are designed for dashboards, forms, and listings, ensuring smooth navigation and usability.

B. Backend API Design

The backend follows RESTful principles and manages data exchange between frontend and database. APIs handle authentication, internship postings, application processing, and administrative verification.

C. Database Structure

A NoSQL database is used to store user profiles, internship listings, application data, and notifications. This structure supports scalability and real-time updates.

D. Security and Authentication

Firestore Authentication ensures secure login and role-based access. Only authorized users can perform actions permitted by their role, ensuring system security and data protection.

XI. COMPARISON WITH EXISTING INTERNSHIP PLATFORMS

To evaluate the effectiveness and uniqueness of the proposed Micro-Internship Portal, it is compared with a widely used and recognized internship platform, *Internshala*. Internshala is one of the leading online platforms in India that provides internship and training opportunities across various domains and industries.

While Internshala offers a large number of internship listings and serves a broad user base, it is primarily designed

as a general-purpose recruitment platform. It focuses on long-duration internships and does not provide institutional integration, faculty verification, or structured guidance specifically tailored for early-year students.

Table III presents a detailed comparison between Internshala and the proposed Micro-Internship Portal.

TABLE III
COMPARISON BETWEEN INTERNSHALA AND PROPOSED MICRO-INTERNSHIP PORTAL

Feature	Internshala	Proposed System
Target Users	Mostly senior students	Early-year students (beginner-focused)
Internship Type	Long-duration internships	Short-term micro-internships
Verification	No academic verification	Faculty-admin verification
Recommendation System	Basic filtering	AI-based scoring + reasoning
Personalization	Limited	Skill-based + AI-enhanced matching
Real-Time Updates	Limited	Real-time (Firestore listeners)
Learning Guidance	Not provided	Skill gap insights + reasoning
Academic Integration	Not available	Fully integrated with institution
System Approach	Job listing platform	Guided learning + internship system

Unlike Internshala, the proposed Micro-Internship Portal is specifically designed as an institute-driven system that addresses the limitations of traditional platforms. The inclusion of faculty verification ensures authenticity and builds trust among students. Furthermore, the integration of an AI-based recommendation system provides match scores, identifies relevant skills, and explains the reasoning behind each recommendation, which enhances decision-making.

The system also supports real-time updates using Firestore listeners, ensuring instant feedback and improved user experience. Overall, the proposed platform offers a more structured, secure, and intelligent solution tailored for early-stage learners and institutional use.

XII. PERFORMANCE EVALUATION AND SYSTEM ANALYSIS

To assess the effectiveness of the proposed Micro-Internship Portal, a qualitative and functional evaluation was conducted based on usability, system responsiveness, security, and workflow efficiency. Since the platform is designed as an academic and institute-driven system, the evaluation focuses on practical usability and operational reliability rather than large-scale commercial metrics.

The integration of AI-based recommendations significantly improved user engagement and decision-making. The system was able to generate meaningful match scores and explanations, even in cases of partial skill overlap. The hybrid

architecture ensured consistent performance through fallback mechanisms.

A. Usability Analysis

The system interface is designed with early-year engineering students in mind, emphasizing simplicity and guided navigation. Students can easily register, update profiles, browse internships, and track applications without requiring prior exposure to professional recruitment platforms. The use of dashboards and categorized listings reduces cognitive load and improves user engagement.

Organizations benefit from a structured posting mechanism where task requirements, skills, and deliverables are clearly defined. Faculty administrators are provided with dedicated moderation interfaces that simplify verification and approval processes. Overall, role-specific interfaces ensure clarity and minimize operational complexity.

B. System Responsiveness and Scalability

The use of a cloud-hosted backend and NoSQL database allows the system to handle concurrent users efficiently. Real-time updates for application status and notifications ensure that students and organizations receive immediate feedback. This is particularly important in micro-internship workflows, where short timelines and quick responses are critical.

Scalability is supported through modular frontend components and RESTful backend services. As the number of users and internship postings increases, the system can scale horizontally without major architectural changes.

C. Security and Reliability Evaluation

Security is enforced through Firebase Authentication and role-based access control. Only verified users can access system functionalities relevant to their role. Administrative verification of organizations and internship postings significantly reduces the risk of fraudulent activities, a common issue in open internship platforms.

Data integrity is maintained through controlled API access and authenticated database operations. These mechanisms ensure reliable system performance and protect sensitive academic and user information.

XIII. AI-BASED RECOMMENDATION ANALYSIS

Table IV presents the AI-based internship matching results obtained from the system.

In practical testing, the backend API initially returned zero recommendations due to strict pre-filtering, as only two approved internships were available and both resulted in 0% match during local scoring. As a result, the AI enhancement step was skipped and the system automatically activated a client-side fallback mechanism.

This behavior demonstrates the robustness of the hybrid recommendation system, where the platform ensures continuous functionality even when server-side AI processing does not

TABLE IV
AI-BASED INTERNSHIP MATCHING RESULTS

Internship	Company	Match	Matched Skills	Reason
Campus Ambassador	Youth Marketer	60%	Communication, Interpersonal Skills	Partial alignment; lacks leadership and marketing skills.
CS Articleship	Akshay B	60%	Python, System Design, Problem Solving	Core skills match partially; missing compliance and communication skills.
Hospital Audit	Value Kare Healthtech LLP	20%	Data Entry, Data Interpretation	Limited overlap with required auditing and domain-specific skills.
Video Editing	Attento Technologies	20%	Python, React	Mismatch with required video editing tools.

yield results. The fallback mechanism performs local scoring combined with AI inference at the client level, ensuring that users always receive meaningful recommendations.

To enhance internship discovery, an AI-based recommendation system was integrated into the Micro-Internship Portal. The system uses a hybrid approach combining rule-based filtering and external AI model inference (Groq API with LLaMA 3.1) to generate match scores, identify relevant skills, and provide reasoning for recommendations.

Initially, internships are filtered using a local pre-scoring mechanism based on skill matching. Selected candidates are then passed to the AI model for deeper evaluation. If no results are returned, a fallback mechanism ensures that recommendations are still generated on the client side.

A. Key Observations

- The AI model successfully identifies partial matches instead of rejecting all internships.
- Skill gaps are clearly highlighted, helping students improve their profiles.
- Backend pre-filtering reduces unnecessary API calls and improves efficiency.
- The fallback mechanism ensures uninterrupted recommendations even when AI is skipped.
- The system uses a hybrid approach combining rule-based filtering and AI reasoning.

XIV. IMPACT ON STAKEHOLDERS AND EDUCATIONAL OUTCOMES

The proposed Micro-Internship Portal positively impacts all key stakeholders involved in the academic and industry ecosystem, including students, organizations, and academic institutions.

A. Impact on Students

For early-year engineering students, the platform provides structured exposure to real-world problem solving at an early stage of their academic journey. By working on short-duration, skill-focused tasks, students gain confidence, improve technical proficiency, and develop professional discipline. This early exposure helps students make informed career decisions and enhances long-term employability.

Additionally, the institute-verified nature of the platform builds trust and reduces uncertainty, enabling students to focus on learning rather than legitimacy concerns.

B. Impact on Organizations

Organizations, especially startups and small enterprises, benefit by gaining access to motivated students for short-term tasks without long-term commitments. The structured workflow allows organizations to clearly define expectations and efficiently evaluate student applications. Faculty verification further enhances trust and accountability, encouraging organizations to participate actively.

C. Impact on Academic Institutions

From an institutional perspective, the platform strengthens industry-academia collaboration and enables faculty members to monitor student engagement beyond the classroom. The system can also support outcome-based education by aligning micro-internship activities with program objectives and skill development goals.

Overall, the platform acts as a bridge between theoretical education and practical industry requirements, enhancing the quality of engineering education.

XV. LIMITATIONS AND ETHICAL CONSIDERATIONS

While the proposed Micro-Internship Portal offers several advantages, certain limitations must be acknowledged. The system currently relies on manual administrative verification, which may increase faculty workload as the number of organizations grows. Automation of preliminary verification using intelligent filtering mechanisms can address this issue in future versions.

Another limitation is the dependency on user-provided data, such as skills and task descriptions, which may vary in accuracy. Proper guidelines and periodic audits are required to maintain data quality and relevance.

From an ethical standpoint, the system is designed to ensure transparency, fairness, and equal access. All internship postings are verified to prevent exploitation and unpaid labor. Student data privacy is protected through secure authentication and controlled access policies. These considerations are critical to maintaining trust and institutional credibility.

XVI. CONCLUSION AND FUTURE WORK

This paper presented a secure, institute-driven Micro-Internship Portal aimed at providing early-year engineering students with structured, verified, and skill-based industry exposure. By integrating faculty verification and role-based workflows, the proposed system addresses limitations of existing internship platforms.

Future enhancements include AI-based internship recommendations, performance analytics dashboards, certificate generation, and mobile application support. These additions can further improve user experience and system effectiveness.

XVII. REFERENCES

- 1) Parker Dewey, "Micro-Internships and Experiential Learning," 2021.
- 2) D. Boud and N. Solomon, *Work-Based Learning: A New Higher Education?*, Open University Press, 2001.
- 3) M. Eraut, "Informal Learning in the Workplace," *Studies in Continuing Education*, 2004.
- 4) Groq Inc., "Groq API Documentation (LLaMA 3.1 Models)," 2024. Available: <https://groq.com/>
- 5) Meta AI, "LLaMA 3: Open and Efficient Foundation Language Models," 2024.
- 6) Firebase, "Firebase Documentation – Authentication, Firestore, and Cloud Functions," 2024. Available: <https://firebase.google.com/docs>
- 7) React.js, "React Documentation," 2024. Available: <https://reactjs.org/docs>
- 8) Node.js Foundation, "Node.js Documentation," 2024. Available: <https://nodejs.org>
- 9) Express.js, "Express Web Framework Documentation," 2024. Available: <https://expressjs.com/>
- 10) Internshala, "Internshala Internship Platform," 2024. Available: <https://internshala.com>
- 11) D. Kolb, *Experiential Learning: Experience as the Source of Learning and Development*, 2nd ed., Prentice Hall, 2014.
- 12) J. Biggs and C. Tang, *Teaching for Quality Learning at University*, 4th ed., McGraw-Hill Education, 2011.
- 13) M. Felder and R. Brent, "Active learning: An introduction," *ASQ Higher Education Brief*, vol. 2, no. 4, pp. 1–5, 2009.
- 14) A. S. Kulkarni et al., "Design and Evaluation of Internship Matching Systems using Recommender Techniques," in *Proc. IEEE ICALT*, 2022, pp. 142–150.